

Rajesh Kumar

Assistant Professor of Physics

SIDO KANHU MURMU UNIVERSITY, DUMKA

Dumka, Jharkhand
India

+91 7352949118

✉ kr.rajesh.phy@gmail.com

🌐 rajeshphy.github.io

Born: 11 April 1995



Research Interests

Quantum Mechanics, Spectral Methods, Computational Physics

Education

- 2016–2018 **M.Sc. Physics, IIT Kanpur**
CGPA: 7.4/10
- 2013–2016 **B.Sc. Physics (Hons.), St. Xavier's College, Ranchi**
80%
- 2011–2013 **Intermediate, Jharkhand Academic Council**
71%
- 2011 **Matriculation, ICSE Board**
88%

Research Publications

- 2024 R. Kumar, R.K. Yadav, A. Khare, **Rationally extended harmonic oscillator potential, isospectral family and the uncertainty relations**, Annals of Physics, 463 (2024): 169623. [DOI](#)
- 2025 R. Kumar, R.K. Yadav, A. Khare, **Rational extension of anisotropic harmonic oscillator potentials in higher dimensions**, Annals of Physics, 479 (2025): 170045. [DOI](#)
- 2025 R.K. Yadav, R. Kumar, A. Khare, **Rational Extension of Quantum Anisotropic Oscillator Potentials with Linear/Quadratic Perturbations**, arXiv:2504.08236 (2025). [arXiv](#)

Research & Professional Experience

- Oct 2022–Present **Assistant Professor, Model College Dumka & SKMU University**
Teaching PG Physics; mentoring computational and theoretical projects.
- Nov 2020–Oct 2022 **Assistant Professor, Deoghar College, SKMU University**
Taught undergraduate and postgraduate physics courses.
- Mar 2019–Mar 2020 **Project Scientist–B, INCOIS, Hyderabad**
Analyzed ocean wave spectra with Python & shell scripts; mesh generation with Gmsh.
- Sep 2018–Feb 2019 **Junior Research Fellow, Nanophotonics Lab, IIT-ISM Dhanbad**
Modeled surface plasmon resonance via ellipsometry (theoretical & computational).

Short Courses

- Sep 2024 **Planetary Science (Online), CSSTEAP (UN-affiliated), PRL Ahmedabad**
Completed short course on planetary science conducted by PRL under CSSTEAP.

Achievements

- 2018 **GATE Physics**
All India Rank 484
- 2017 **CSIR-JRF (Physics)**
All India Rank 218

2016 **National Graduate Physics Examination (NGPE)**

State Topper, Bihar & Jharkhand (Conducted by IAPT)

2011, 2013 **Block Topper**

Class 10 & 12, Silli, Ranchi

Technical Skills

Programming: Python, C, C++, Java, Shell scripting, LaTeX.

Web Development: Django, HTML, CSS, JS, Jekyll.

Tools: Gmsh, Tkinter GUI, Git.